

Total Marks - 84**Attempt Questions 1 – 7****All questions are equal value.**

Answer each question in a SEPARATE writing booklet.

Question 1 (12 Marks) Use a SEPARATE writing booklet.**Marks**

- (a) Find the exact value of $\int_0^{\frac{\pi}{6}} \sec^2 2x dx$. 2
- (b) Solve the inequation $\frac{x}{x-3} \geq 2$, $x \neq 3$. 3
- (c) Find the acute angle between the lines $x - y + 3 = 0$ and $2x + y + 1 = 0$.
Give your answer correct to the nearest minute. 2
- (d) Find the co-ordinates of P, the point which divides the interval A(−5, −3) and B(4, −6) externally in the ratio 2 : 3. 2
- (e) When $P(x) = x^3 + 3x^2 - mx + n$ is divided by $(x + 2)$ the remainder is 9 and when $P(x)$ is divided by $(x - 3)$ the remainder is 49.
Find the values of m and n . 3

Question 2 (12 Marks) Use a SEPARATE writing booklet.

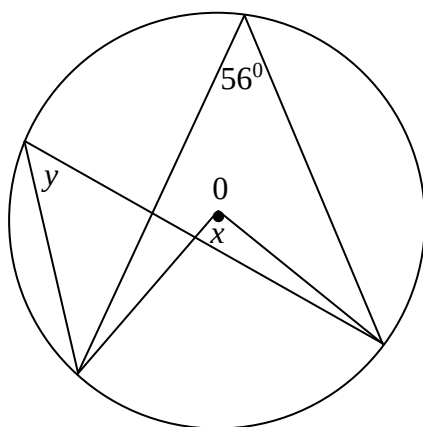
- (a) Sketch the graph of $y = \cos^{-1}\left(\frac{x}{2}\right)$. 2
Your graph must indicate the domain and range.
- (b) Find $\frac{d}{dx}(x \sin^{-1} 5x)$ 2
- (c) (i) Write $2 \cos \theta + \sin \theta$ in the form $A \cos(\theta - \alpha)$ where $A > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. 2
- (ii) Hence, or otherwise solve the equation $2 \cos \theta + \sin \theta = \sqrt{5}$ for $0 \leq \theta \leq 2\pi$. 2

Give your answer correct to 2 decimal places.

Question 2 Continued.**Marks**

(d) Find the general solution for $\cos \theta = \frac{\sqrt{3}}{2}$. 2

- (e) O is the centre of the circle. Find the value of the pronumerals, x and y , giving reasons. 2

**Question 3 (12 Marks)** Use a SEPARATE writing booklet.

- (a) Use the Binomial Theorem to find the term independent of x in the expansion of $\left(x^2 - \frac{1}{2x}\right)^{18}$. 3

- (b) Given $f(x) = \cos x - \ln x$

- (i) Show that a root of $f(x) = 0$ lies between 0.5 and 1.5. 1

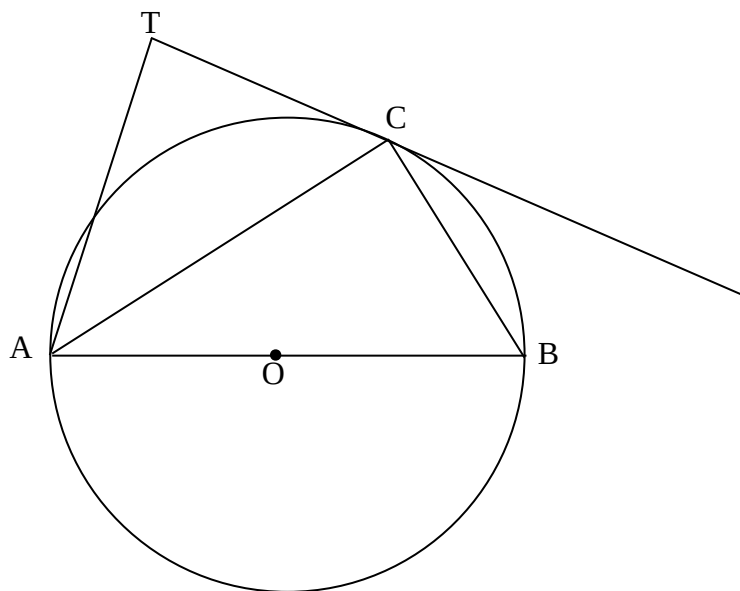
- (ii) Use Newton's Method once to find an approximation to this root of the equation, correct to 2 decimal places, starting with $x = 1$. 2

(c) Evaluate $\int_0^{0.4} \frac{3dx}{4 + 25x^2}$ 3

Questions continues on Page 3.

Question 3 Continued.**Marks**

- (d) AOB is the diameter of a circle, centre O. C is the point of contact of the tangent, TC such that AC bisects $\angle TAB$. Prove that AT is perpendicular to TC. **3**



Question 4 (12 Marks) Use a SEPARATE writing booklet.

- (a) Show by Mathematical Induction that $9^n - 7^n$ is divisible by 2, for $n \geq 1$. **4**
- (b) How many different arrangements of the letters of the word NEWINGTON are possible? **2**
- (c) (i) A committee of eight people is to be formed from a group of twenty people. In how many different ways can the committee be formed? **1**
- (ii) This group consists of twelve men and eight women, how many ways can committee of four men and four women be formed if Kate and Pete must be included? **2**
- (iii) In how many ways can these 8 committee members sit around a circular table with no conditions restricting where anyone sits? **1**
- (iv) If the men and women alternate where they sit, how many arrangements are possible? **2**

Question 5 (12 Marks) Use a SEPARATE writing booklet.**Marks**

- (a) P $(2ap, ap^2)$ and Q $(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$.

The tangents at P and Q meet at T which is always on the parabola $x^2 = -4ay$.

- (i) Given the equation of the tangent at P is $y - px + ap^2 = 0$. 2

Show that T is the point $(a(p + q), apq)$.

- (ii) Show that $p^2 + q^2 = -6pq$. 1

- (iii) Find M, the midpoint of PQ. 1

- (iv) Find the equation of the locus of M. 3

- (b) A casserole is cooling in a room that has a constant temperature of C . At a given time, t minutes the temperature decreases according to $\frac{dT}{dt} = -k(T - 22)$ where k is a positive constant.

- (i) Show that $T = 22 + Ae^{-kt}$ is a solution to the equation. 1

- (ii) Given that the initial temperature of the casserole is 80°C and it cools to 60°C after 10 minutes. Find A and k . 2

- (iii) How long will it take for the casserole to cool to 30°C ? 2

Question 6 (12 Marks) Use a SEPARATE writing booklet.

- (a) Ben fires an arrow horizontally with a speed of 60 ms^{-1} from the top of a 20 m high cliff. Use $g = 10 \text{ m/s}^2$.
- (i) Show that the position of the arrow at time, t seconds is given by $x = 60t$ and $y = 20 - 5t^2$. 2
- (ii) Find the time taken for the arrow to hit the ground. 1
- (iii) Find the distance the arrow is from the base of the cliff when it hits the ground. 1
- (iv) Find the acute angle to the horizontal at which the arrow hits the ground. 2
- (b) A particle's motion is defined by the equation, $v^2 = 12 + 4x - x^2$ where x is its displacement from the origin in metres and v is its velocity in ms^{-1} . Initially, it is 6 metres to the right of the origin.
- (i) Show that the particle is moving in Simple Harmonic Motion. 1
- (ii) Find the centre, the period and the amplitude of the motion. 3
- (iii) The displacement of the particle at any time is given by the equation $x = a \sin(nt + \theta) + b$. 2
- Find the values of, θ and b given $0 \leq \theta \leq 2\pi$.

Question 7 (12 Marks) Use a SEPARATE writing booklet.

(a) Using the substitution $u = 2x + 1$ find $\int_0^1 \frac{4x}{2x+1} dx$. **4**

(b) For $(a + b)^n$ the general term is $T_{k+1} = {}^nC_k a^{n-k} b^k$.

For the following expansion $(3 + 11x)^{19}$:

(i) Show that $\frac{T_{k+1}}{T_k} = \frac{11x(20 - k)}{3k}$. **2**

(ii) Hence find the greatest coefficient of $(3 + 11x)^{19}$. **3**

(c) (i) Use the binomial theorem to obtain an expansion for $(1 + x)^{2n} + (1 - x)^{2n}$ **1**
where n is a positive integer.

(ii) Hence evaluate $1 + {}^{20}C_2 + {}^{20}C_4 + \dots + {}^{20}C_{20}$. **2**

END OF PAPER

① a) $\int_0^{\pi/6} \sec^2 2x dx = \left[\frac{\tan 2x}{2} \right]_0^{\pi/6} = \frac{\tan \pi/3 - \tan 0}{2} = \frac{\sqrt{3}}{2}$ ① Correct Answer

b) $\frac{x}{x-3} \geq 2 \quad \times (x-3)^2 (x-3) \geq 2(x-3)^2$

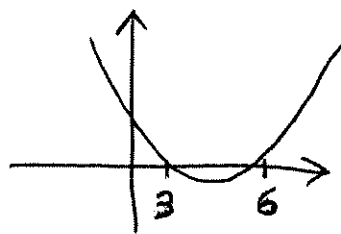
$x^2 - 3x \geq 2x^2 - 12x + 18$ ①

$0 \geq x^2 - 9x + 18$

$0 \geq (x-6)(x-3)$ ①

$3 < x \leq 6$

① Answer



c) $\tan \alpha = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$x - y + 3 = 0$

$y = x + 3 \quad m_1 = 1$

$2x + y + 1 = 0$

$y = -2x - 1 \quad m_2 = -2$ ①

$\tan \alpha = \left| \frac{1 + 2}{1 - 2} \right| = 3$ ①

$\alpha = 71^\circ 33' 54'' \dots = 71^\circ 34' \text{ (nearest minute)}$

d) A(-5, -3) B(4, -6) External Division 2:-3

$xc = \frac{mx_2 + nx_1}{m+n} = \frac{8 + 15}{-1} = -23$ ① $y = \frac{my_2 + ny_1}{m+n} = \frac{(2 \times -6) + (-3 \times -3)}{2-3} = \frac{-3}{-1} = 3$

P has co-ordinates $(-23, 3)$. ①

e) $P(x) = x^3 + 3x^2 - mx + n$

$P(-2) = 9$

$-8 + 12 + 2m + n = 9$

$2m + n = 5$ ① ①

$P(3) = 49$

$27 + 27 - 3m + n = 49$

$-3m + n = -5$ ② ①

① - ② $5m = 10 \quad \therefore m = 2 \quad \therefore n = 1$. ①

MARKERS COMMENTS

a) Generally well done. Some gave coefficient of 2 not $\frac{1}{2}$

b) Most students knew to multiply by square of denominator. Mistakes came collecting terms, factoring & interpreting region. Students were not penalised for failure to recognise $x \neq 3$.

c) Many recalled the formula incorrectly

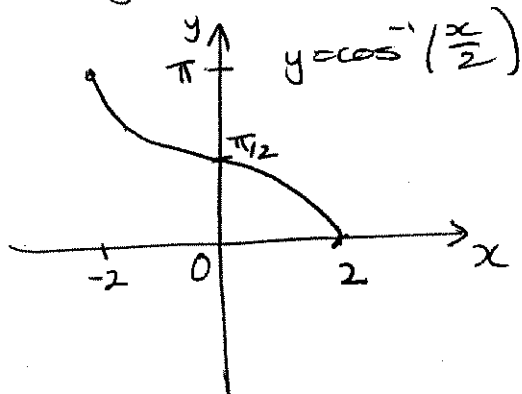
d) Students were poor at applying standard formula. Most common error was $mx_1 + nx_2$ in numerator.

e) Students recognising factor theorem generally completed this question well.

(a) sketch $y = \cos^{-1}\left(\frac{x}{2}\right)$

Domain $-1 \leq \frac{x}{2} \leq 1$
 $-2 \leq x \leq 2$

Range $0 \leq y \leq \pi$



1 mark correct curve
 1 mark correct domain
 and range.

(b) $\frac{d}{dx} (x \sin^{-1} 5x)$

$$= \sin^{-1} 5x + x \times \frac{5}{\sqrt{1-25x^2}}$$

$$= \sin^{-1} 5x + \frac{5x}{\sqrt{1-25x^2}}$$

1 mark correct differentiation
 of $\sin^{-1} 5x$

1 mark correct answer
 from their working

(c) $2\cos\theta + \sin\theta = A\cos(\theta - \alpha)$ $A > 0$

$$0 \leq \alpha \leq \frac{\pi}{2}$$

1 mark $A = \sqrt{5}$

$$A = \sqrt{2^2 + 1^2}$$

$$A = \sqrt{5}$$

OR

$$\therefore 2\cos\theta + \sin\theta = \sqrt{5}\cos(\theta - \alpha)$$

and $\tan\alpha = \frac{1}{2}$

$$\alpha = 0.463647609^\circ$$

$$\alpha = 0.46^\circ \text{ (2dp)}$$

1 mark $\alpha = \tan^{-1}\left(\frac{1}{2}\right)$
 or $\alpha = 0.463647\dots$
 $\alpha = \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$
 $\alpha = \sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$

$\therefore 2\cos\theta + \sin\theta = \sqrt{5}\cos(\theta - 0.46^\circ)$ 1 mark correct solution.

OR accept $= \sqrt{5}\cos(\theta - \alpha)$ $\alpha = \tan^{-1}\left(\frac{1}{2}\right)$

degrees not accepted.

$$(c)(ii) 2\cos\theta + \sin\theta = \sqrt{5}$$

$$\therefore \sqrt{5}\cos(\theta - 0.46^\circ) = \sqrt{5}$$

$$\therefore \cos(\theta - 0.46^\circ) = 1$$

1 mark correct equation

$$\therefore \theta - 0.46^\circ = 0, 2\pi, 4\pi, \dots$$

$$\text{But } 0 \leq \theta \leq 2\pi$$

$$\theta = 0.46^\circ \text{ (2dp)}$$

1 mark correct θ

$$(d) \cos\theta = \frac{\sqrt{3}}{2}$$

$$\cos\theta = \cos\frac{\pi}{6}$$

general solution

$$\theta = 2n\pi \pm \alpha$$

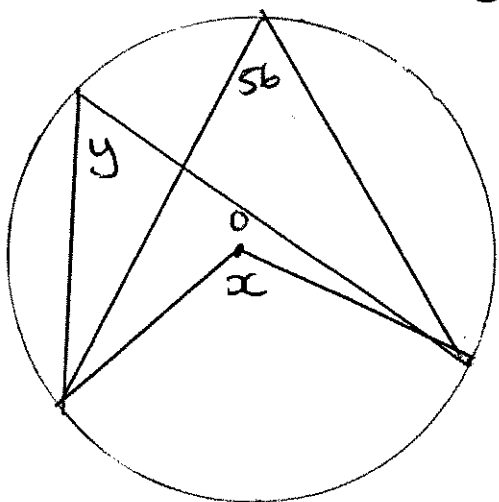
$$\theta = 2n\pi \pm \frac{\pi}{6}$$

$$\text{OR } \theta = -\frac{\pi}{6}, \frac{\pi}{6}, \frac{13\pi}{6}, \frac{23\pi}{6}, \dots$$

1 mark for $\frac{\pi}{6}$

1 mark correct answer only

(e)



$$x = 2 \times 56^\circ \text{ (angle at the centre and the circumference)}$$

$$x = 112^\circ$$

$$y = 56^\circ \text{ (angles on the same arc)}$$

(angles on the same chord)
(angles in the same segment)

1 mark each
correct value with correct
reason.

Question 3

(a) Independent term $\Rightarrow x^0$

$$\begin{aligned}
 (x^2 - \frac{1}{2}x^{-1})^{18} &\Rightarrow {}^{18}C_r (x^2)^{18-r} (-\frac{1}{2})^r (x^{-r}) \\
 &= {}^{18}C_r x^{36-2r} x^{-r} (-\frac{1}{2})^r \\
 &= {}^{18}C_r (-\frac{1}{2})^r x^{36-3r}
 \end{aligned}$$

$$\therefore 36 - 3r = 0$$

$$\therefore r = 12$$

$$\therefore \text{Independent term} = {}^{18}C_{12} \left(\frac{1}{2}\right)^{12}$$

$$= \frac{4641}{1024}$$

- { 3 mks correct ans.
 2 mks correct method for finding r
 1 mk correct expansion

(b) (i) For $f(x) = \cos x - \ln x$ [1 mk] $f(x)$ is $\downarrow$ for $0.5 \leq x \leq 1.5$

$$f(0.5) > 0 \quad \text{ie.} \quad f(0.5) \approx 1.57$$

$$f(1.5) < 0 \quad \text{ie.} \quad f(1.5) \approx -0.33$$

 $\therefore$ root lies in the interval $0.5 \leq x \leq 1.5$

(ii) By Newton's Method:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \quad ; \text{ if } x_0 = 1$$

then $x_1 = 1 - \frac{f(1)}{f'(1)}$

$$f'(x) = -\sin x - \frac{1}{x} \quad \therefore f'(1) = -\sin(1) - 1$$

$$f(x) = \cos x - \ln x \quad \therefore f(1) = \cos 1$$

$$\therefore x_1 = 1 + \frac{\cos 1}{\sin(1) + 1}$$

$$\therefore x_1 \approx 1.29 \quad (2 \text{ dp})$$

1 mk
 { correct use of
 formula
 gives correct ans

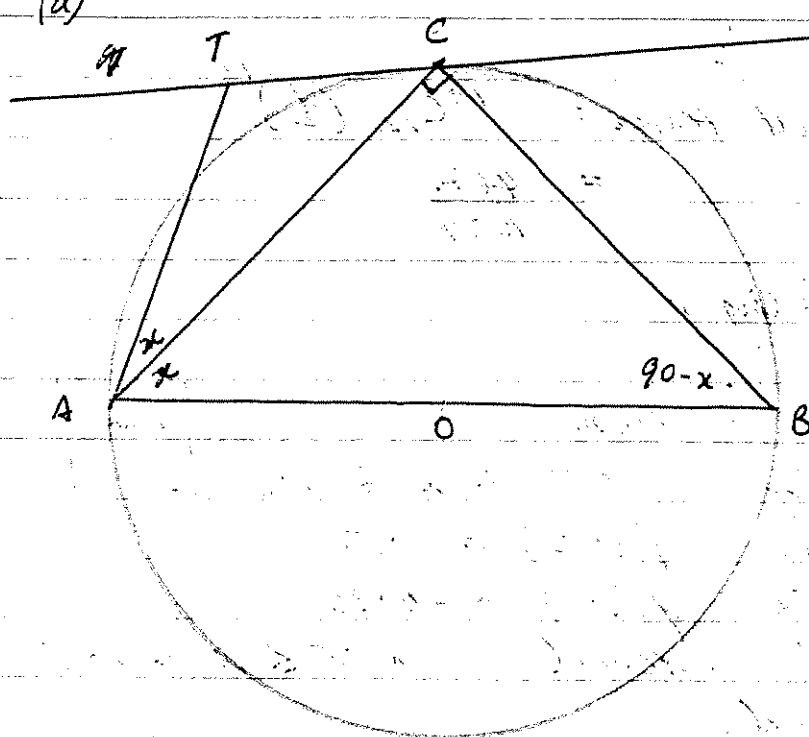
$$(c) \int_0^{0.4} \frac{3 \, dx}{4 + 25x^2} = 3 \int_0^{0.4} \frac{1}{4 + 25x^2} \, dx$$

$$= \frac{3}{25} \left[\frac{25}{2} \cdot \tan^{-1} \frac{5x}{2} \right]_0^{0.4}$$

$$\left[\begin{array}{l} 3 \text{ mks correct ans} \\ 2 \text{ mks correct integral} \\ 1 \text{ mks correct integration method} \end{array} \right] = 0.3 \left[\tan^{-1}(1) - \tan^{-1}(0) \right]$$

$$= \frac{3\pi}{40}$$

(d)



TC is a tangent

AC bisects $\angle TAB$.

AOB is a diameter

$\angle ACB = 90^\circ$ (angle in a semi-circle).

$\therefore \angle ABC = 90 - x$.

$\therefore \angle TCA = 90 - x$ (angle in alternate segment)

$\therefore \angle ATC = 90^\circ$ (angle sum of $\triangle ATC$)

$\left[\begin{array}{l} 3 \text{ mks correct method} \\ 2 \text{ mks 1 mks (angle in semi)} \\ 1 \text{ mks (angle in alt seg)} \\ 1 \text{ mks (angles in } \triangle) \end{array} \right]$

Question 4

a) Step 1 ~~Let~~ Prove true for $n=1$

$$9^1 - 7^1 = 2 \therefore \text{divisible by } 2$$

Step 2 Assume true for $n=k$

$$9^k - 7^k = 2M \text{ where } M \text{ is positive integer}$$

Step 3 Prove true for $n=k+1$

$$9^{k+1} - 7^{k+1} = 9 \cdot 9^k - 7 \cdot 7^k \quad \checkmark$$

$$= 7(9^k - 7^k) + 2 \cdot 9^k \quad \checkmark$$

$$= 7(2M) + 2 \cdot 9^k \text{ (From Step 2)}$$

$$= 2(7M + 9^k) \quad \checkmark$$

$\therefore 9^{k+1} - 7^{k+1}$ is divisible by 2

Step 4

b) $\frac{9!}{3!} = 60480$

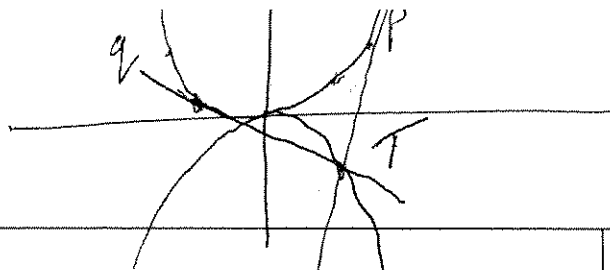
c) i) ${}^{20}C_8 = 125970$

ii) ${}^{11}C_3 \times {}^7C_3 = 5775$

iii) $7! = 5040$

iv)

$$4! \times 3! = 144$$



5a)		Marking Criteria
	(i) As the tangent at $p(2ap, ap^2)$ is $y - px + ap^2 = 0$ (1), therefore the tangent at Q is $y - qx + aq^2 = 0$ (2). Solving (1) and (2) simultaneously: $(2) - (1)$ $(p - q)x - a(p^2 - q^2) = 0$ $\therefore x = \frac{a(p^2 - q^2)}{(p - q)}$ $= a(p + q)$ $\therefore y = p \times a(p + q) - ap^2$ $= apq$ $\therefore T(a(p + q), apq)$ (ii) As T lies on $x^2 = -4ay$ $\therefore (a(p + q))^2 = -4a \times apq$ $\therefore p^2 + 2pq + q^2 = -4pq$ $\therefore p^2 + q^2 = -6pq$ (iii) Mid-point of PQ: $M\left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2}\right) = M\left(a(p + q), \frac{a}{2}(p^2 + q^2)\right)$ (iv) Let $x = a(p + q), \quad y = \frac{a}{2}(p^2 + q^2)$ $\therefore y = \frac{a}{2}((p + q)^2 - 2pq)$ $\therefore y = \frac{a}{2}\left(\left(\frac{x}{a}\right)^2 - 2pq\right)$ $\therefore y = \frac{x^2}{2a} - apq$ $\therefore y = \frac{x^2}{2a} + \frac{a(p^2 + q^2)}{6}$ $\therefore y = \frac{x^2}{2a} + \frac{a(p^2 + q^2)}{2} \times \frac{1}{3}$ $\therefore y - \frac{y}{3} = \frac{x^2}{2a}$ $\therefore x^2 = \frac{4a}{3}y$ Locus of M is a parabola.	Marking Criteria 1 mark process 1 mark for correct working 1 mark for correctly using given information 1 mark for correct working 1 mark for starting with the correct parameters! 1 mark for correct process 1 mark for correct answer

5b)		Marking Criteria
	Given $\frac{dT}{dt} = -k(T - 22)$ (i) $T = 22 + Ae^{-kt}$ $\frac{dT}{dt} = -kAe^{-kt}$ $\frac{dT}{dt} = -k(T - 22) \quad \text{as} \quad Ae^{-kt} = T - 22$ (ii) Now, when $t = 0$, $T = 80^\circ$ $\therefore 80 = 22 + A \times 1$ $\therefore A = 58$ And when $t = 10$, $T = 60^\circ$ $\therefore 60 = 22 + 58 \times e^{-k \times 10}$ $\therefore e^{-10k} = \frac{19}{29}$ $\therefore k = -\frac{1}{10} \ln \left(\frac{19}{29} \right)$ (iii) As $T = 22 + 58 e^{\frac{t}{10} \left(\ln \frac{19}{29} \right)}$ when $T = 30$ $\frac{4}{29} = e^{\frac{t}{10} \left(\ln \frac{19}{29} \right)}$ $\therefore \frac{t}{10} = \frac{\ln \left(\frac{4}{29} \right)}{\ln \frac{19}{29}}$ $\therefore t \approx 46.84 \text{ min} \quad \left(\begin{array}{l} 46 \text{ min} \\ 51 \text{ sec} \end{array} \right)$	1 mark for correct working 1 mark for correct value of A 1 mark for correct value of k 1 mark for correct process 1 mark for correct answer

Q6

(i) Horizontally

$$\ddot{x} = 0$$

$$\dot{x} = t + c$$

$$\text{when } t=0, \dot{x}=60$$

$$\therefore \dot{x} = 60$$

$$\therefore x = 60t + c$$

$$\text{when } x=0, t=0$$

$$\therefore \boxed{x = 60t}$$

Vertically

$$\ddot{y} = -g = -10$$

$$\dot{y} = -10t + c$$

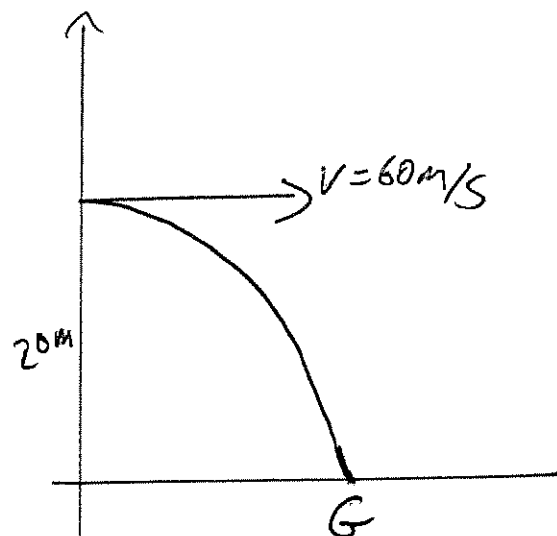
$$\text{when } t=0, \dot{y}=0$$

$$\therefore \dot{y} = -10t$$

$$\therefore y = -\frac{10t^2}{2} + c$$

$$\text{when } t=0, y=20$$

$$\therefore \boxed{y = 20 - 5t^2}$$



[2]

(ii) Arrow hits the ground when $y=0$

$$\therefore 20 - 5t^2 = 0$$

$$(2-t)(2+t) = 0$$

$$\therefore \text{after 2 seconds } [t \geq 0] \quad \checkmark$$

[1]

(iii) when $t=2$

$$x = 60 \times 2 = 120 \text{ m}$$

✓

[1]

(iv) at G

$$\dot{x} = 60 \text{ m/s}$$

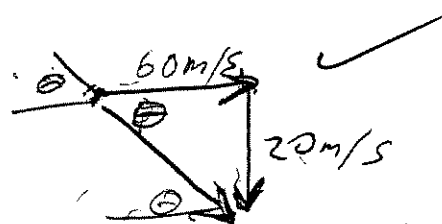
$$\dot{y} = -20 \text{ m/s}$$

$$\tan \theta = \frac{20}{60}$$

$$\theta = \tan^{-1}\left(\frac{1}{3}\right)$$

$$\text{accept } \theta = 18^\circ 26'$$

[2]



(b)

$$v^2 = 12 + 4x - x^2$$

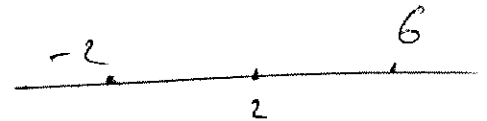
$$(i) \quad \frac{1}{2} v^2 = 6 + 2x - \frac{1}{2} x^2 \quad (1)$$

$$\frac{d(\frac{1}{2} v^2)}{dx} = 2 - x \quad \therefore \text{SHM as } \checkmark$$

$\ddot{x}$ is directly proportional to its displacement.

$$(ii) \quad v^2 = (6 - x)(2 + x)$$

$$v = \pm \sqrt{(6 - x)(2 + x)}$$



Centre of Motion: $x = 2$

$$\text{Period} = 2\pi = 2\pi \text{ sec}$$

$$\text{Amplitude} = 4 \text{ m}$$

✓
✓
✓
(3)

$$(iii) \quad x = a \sin(n\omega t + \theta) + b$$

$$x = 6, a = 4, n = 1, t = 0$$

$$6 = 4 \sin \theta + b$$

$$\text{now } \dot{x} = a n \omega (n\omega t + \theta)$$

$$\dot{x} = 0 \text{ when } t = 0$$

$$\therefore \cos \theta = 0$$

$$\theta = \frac{\pi}{2} \neq$$

(2)

$$\therefore 6 = 4 \sin \frac{\pi}{2} + b \quad \text{when } t = 0, x = 6$$

$$\therefore b = 2 \neq$$

Question:

$$(a) \int_0^1 \frac{4x}{2x+1} dx$$

$$\text{let } u = 2x+1$$

$$\therefore u-1 = 2x$$

$$\therefore \int_0^1 \frac{2x}{2x+1} \cdot 2 dx$$

$$1 du = 2 dx$$

$$\text{when } x=1 \quad u=3$$

$$x=0 \quad u=1$$

$$= \int_1^3 \frac{u-1}{u} \cdot du$$

$$= \int_1^3 \frac{u}{u} - \frac{1}{u} \cdot du$$

$$= \int_1^3 1 - \frac{1}{u} du$$

$$= u - \ln u \Big|_1^3$$

$$= (3 - \ln 3) - (1 - \ln 1)$$

$$= 3 - \ln 3 - 1$$

$$= 2 - \ln 3.$$

① mark

for correct manipulation.

① mark

correct substitution of their variables

① mark

correct integration of their expression as long as the expression is not easier.

① mark

correct answer only.

$$(b) (3+11x)^{19} = (a+b)^n$$

$$\text{then } T_{k+1} = {}^nC_k a^{n-k} \cdot b^k$$

$$(1) \text{ Show } \frac{T_{k+1}}{T_k} = \frac{11x(20-k)}{3k}$$

Using the rule

$$\frac{T_{k+1}}{T_k} = \frac{n-k+1}{k} \cdot \frac{b}{a}$$

$$= \frac{19-k+1}{k} \cdot \frac{(11x)}{3}$$

① mark
correct substitution
nb formula.

$$\therefore \frac{T_{k+1}}{T_k} = \frac{11x(20-k)}{3k}.$$

① mark
correct manipulation

Question 7

$$cc) (1+x)^{2n} + (1-x)^{2n} \quad n > 0$$

$$(1+x)^{2n} = \binom{2n}{0} + \binom{2n}{1}x + \binom{2n}{2}x^2 + \dots + x^{2n}$$

$$(1-x)^{2n} = \binom{2n}{0} - \binom{2n}{1}x + \binom{2n}{2}x^2 + \dots + (-x)^{2n}$$

$$\begin{aligned}(1+x)^{2n} + (1-x)^{2n} &= 2 \times \binom{2n}{0} + 2 \times \binom{2n}{2}x^2 + \dots + 2 \binom{2n}{2n}x^{2n} \\ &= 2 \left(\binom{2n}{0} + \binom{2n}{2}x^2 + \dots + \binom{2n}{2n}x^{2n} \right)\end{aligned}$$

① mark correct answer in any form.

Evaluate

$$1 + {}^{20}C_2 + {}^{20}C_4 + \dots + {}^{20}C_{20}$$

$$= 1 \left(1^{2n} + \binom{2n}{2}x^2 + \dots + \binom{2n}{2n}x^{2n} \right)$$

where $n=10$ $x=1$ ← ① mark correct substitution

$$= 1 \left(1^{20} + {}^{20}C_2 1^2 + {}^{20}C_4 1^4 + \dots + {}^{20}C_{20} 1^{20} \right)$$

$$= 1 \left(1 + {}^{20}C_2 + {}^{20}C_4 + \dots + {}^{20}C_{20} \right)$$

$$= \frac{(1+x)^{2n} + (1-x)^{2n}}{2} \quad \text{where } x=1 \quad n=10$$

$$= \frac{2^{20} + 0^{20}}{2}$$

$$= 2^{19}$$

① mark correct answer
Must show where the answer is derived from using (i)

$$= 524288$$

(i) without the rule : Using $T_{k+1} = {}^nC_k a^{n-k} \cdot b^k$
 $T_k = {}^nC_{k-1} a^{n-(k-1)} \cdot b^{k-1}$
 $T_k = {}^nC_{k-1} a^{n-k+1} \cdot b^{k-1}$

$$T_{k+1} = {}^{19}C_k 3^{19-k} \cdot (11x)^k$$

$$T_k = {}^{19}C_{k-1} \cdot 3^{19-k+1} \cdot (11x)^{k-1}$$

$$= {}^{19}C_{k-1} \cdot 3^{20-k} \cdot (11x)^{k-1}$$

① mark correct expression for T_k

$$\therefore \frac{T_{k+1}}{T_k} = \frac{{}^{19}C_k \cdot 3^{19-k} \cdot (11x)^k}{{}^{19}C_{k-1} \cdot 3^{20-k} \cdot (11x)^{k-1}}$$

$$= \frac{19!}{k! (19-k)!} \cdot 3^{19-k} \cdot (11x)^k$$

$$\frac{19!}{(k-1)! (20-k)!} \cdot 3^{20-k} \cdot (11x)^{k-1}$$

$$= \frac{(k-1)! (20-k)! \cdot 11x}{k! (19-k)! \cdot 3}$$

$$= \frac{(20-k) \cdot 11x}{k \cdot 3}$$

$$= \frac{11x(20-k)}{3k}$$

Working must be shown.

(ii) Greatest coefficient occurs when $\frac{T_{k+1}}{T_k} \geq 1$

$$\therefore \frac{11(20-k)}{3k} \geq 1$$

① mark for their result from (i) without the x .

$$11(20-k) \geq 3k$$

$$220 - 11k \geq 3k$$

$$220 \geq 14k$$

$$14k \leq 220$$

$$k \leq 15\frac{5}{7}$$

$$\therefore k = 15$$

$$\therefore T_{k+1} > T_k \text{ when } k \leq 15\frac{5}{7}$$

$$\therefore T_{16} > T_{15}$$

$$\therefore T_{16} \text{ has the greatest coefficient } T_{16} = {}^{19}C_{15} \cdot 3^4 \cdot 11^{15}$$

① mark correct answer.